

The empirical recurrence for OEIS A316812, recovered from its tail

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Abstract

OEIS A316812 records “Empirical recurrence of order 62” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 123$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 64$. Its last coefficient is $c_{62} = 630884 \neq 0$, so no further tail satisfies anything shorter; any order-62 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A316812 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 0, 1, 2, 3, 4, 5 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 512, 11339, 246196, 5391628, 117897523, 2578023456, 56382210217, ...

The entry states:

Empirical recurrence of order 62 (see link above).

As of the “Last modified” line on the live entry (Apr 11 2022 18:19:16, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 123$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 123$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 62: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 62.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 246$ exact terms from $n = 2$ on, it returns order 62 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	17	c_{17}	15471745114	c_{33}	−12718392049868	c_{49}	220878410866
c_2	169	c_{18}	49334476046	c_{34}	−4872581401350	c_{50}	−89130594676
c_3	−530	c_{19}	52922344001	c_{35}	−327616756933	c_{51}	2329866772
c_4	−16879	c_{20}	20064702398	c_{36}	−1452831998581	c_{52}	47356438071
c_5	−52764	c_{21}	−143597726236	c_{37}	4010000408982	c_{53}	−6147767101
c_6	331395	c_{22}	−473762539993	c_{38}	7145334129377	c_{54}	−821345432
c_7	2977251	c_{23}	−384428908645	c_{39}	951835295815	c_{55}	12387149836
c_8	3888311	c_{24}	887364856765	c_{40}	−2591465662137	c_{56}	4402930342
c_9	−36187084	c_{25}	2573886063629	c_{41}	149360890007	c_{57}	−1883132102
c_{10}	−143444119	c_{26}	1143391085224	c_{42}	839301006199	c_{58}	−1410900482
c_{11}	−4103103	c_{27}	−4891853360219	c_{43}	−831507724478	c_{59}	−259914979
c_{12}	1066290475	c_{28}	−7980837971064	c_{44}	−592471394225	c_{60}	95136780
c_{13}	2119199829	c_{29}	546739100758	c_{45}	111004180804	c_{61}	26188470
c_{14}	−1650847025	c_{30}	13198520510418	c_{46}	−284598552690	c_{62}	630884
c_{15}	−11709537837	c_{31}	11163113398899	c_{47}	−285071703389		
c_{16}	−12085758567	c_{32}	−5097734535290	c_{48}	310159627529		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 64$, and it is the recurrence OEIS A316812 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{62} - c_1 x^{62-1} - \dots - c_{62}$. Its constant term is $-c_{62} = -630884$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and

the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 62 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 62, so $\deg q = 62$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 62, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 62, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 630884.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-62 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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